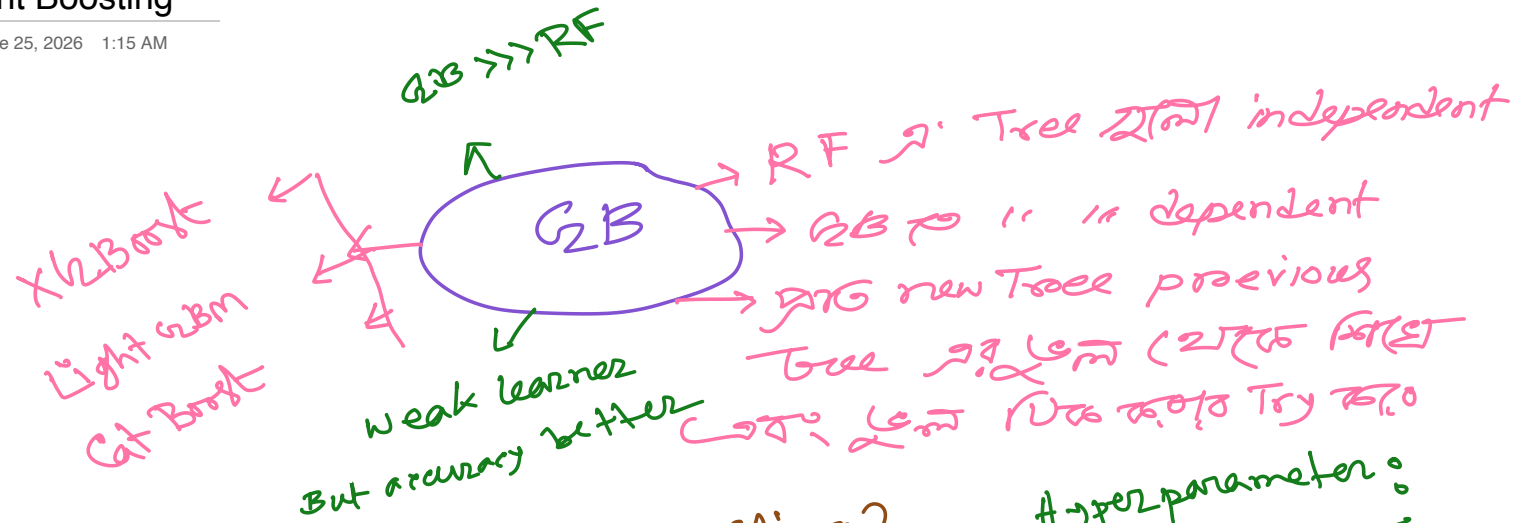


# Gradient Boosting

Thursday, June 25, 2026 1:15 AM



XGBoost: (Extreme Gradient Boosting)

- very popular now.
- Faster & Regularization
- High accuracy

LightGBM:

- very fast
- large dataset & less memory

CatBoost:

- categorical data can handle it.
- Encoding not matter.

Hyperparameter:

- ①  $n\_estimators = \text{num of Tree}$
- ②  $max\_depth$
- ③  $Learning\_rate^{***}$ 
  - Low → More Tree need
  - Better Generalization
  - High → Fast Training
  - overfitting risk

